

Ribbon Classification of Links With 10-14 Crossings

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Thank You



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Overview

1. Definitions
2. A Marvelous Theorem
3. Ribbon Disks
4. Results

What is a Knot?

Definition

A knot is a smooth embedding of S^1 into $\mathbb{R}^3$.

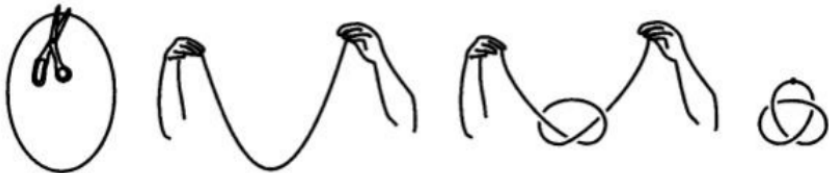


Figure: Constructing a Knot

Visualization of Knots

In order to visualize knots we consider some map $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined as $f(x, y, z) = (x, y)$. This is called the projection of a knot K (but we typically don't distinguish between this projection and the knot itself)

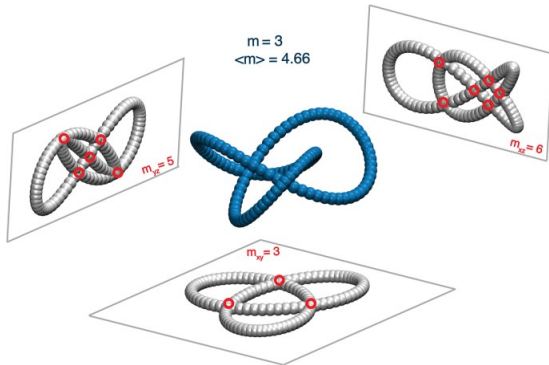


Figure: [Vrgras-Lara, 2017]

Visualization of Knots Cont.



A Notion of 'Sameness'

Definition

Two knots are ambient isotopic to one another if there is a way to continuously deform one to the other without the knot crossing through itself.^a This defines an equivalence relation on the set of all knots.

^aTechnically we are deforming the space they are embedded in



Figure: Two Isotopic Knots

Surfaces

Definition

A surface (2-manifold) is an object such that every point on the object has a neighborhood in the object that is (homeomorphic to) an open disk ^a [Adams, 1994]

^aWe consider surfaces to be embedded (or immersed) in $\mathbb{R}^3$ for the purposes of this talk, but the definition does not require this

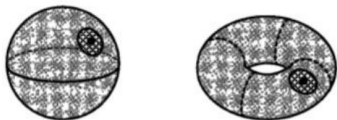


Figure 4.2 Each point on a surface is surrounded by a disk.

Figure: Two Surfaces

Illustrative Anti-Example

The following figures are NOT surfaces

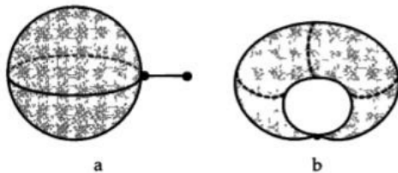


Figure: These are not surfaces

Surface With Boundary

A surface with boundary is created by taking a surface and removing some number of open disks



Figure: Surfaces With Boundary

WAIT!!!!

If we remove open disks can't we find points along the boundary where they don't have an open disk as a neighborhood?

Surfaces with boundary are actually a separate object from surfaces, and they can have neighborhoods homeomorphic to a half disk

Why is This Relevant?

We can consider knots as the boundary of surfaces.



Figure: Trefoil as the Boundary of A Mobius Band

Why is This Relevant? Cont.

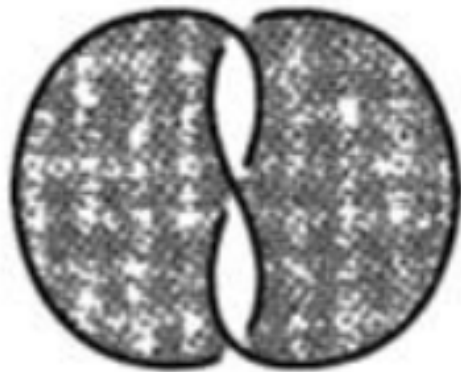


Figure: Trefoil as the Boundary of A Torus

Marvelous Theorem

Theorem (Seifert, 1934)

Given any knot, one can construct an orientable surface with one boundary component which is the knot. Not only is this possible, but there is an algorithm to do this.

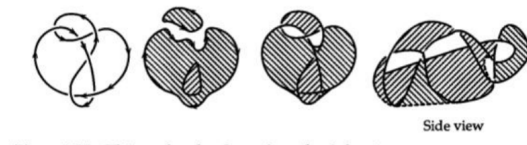


Figure: Seifert Surface for 6_3

This tells us that not only can we embed surfaces with knotted boundary, but we can do so to get any knot, i.e. every knot arises as the boundary of some Seifert surface!!

[Adams, 1994]

Modular Surfaces

Seifert surfaces are nice conceptually but not necessarily the best to work with. One trick is to create a modular surface by 'pushing' the surface down onto the plane, similar to our projections of knots.

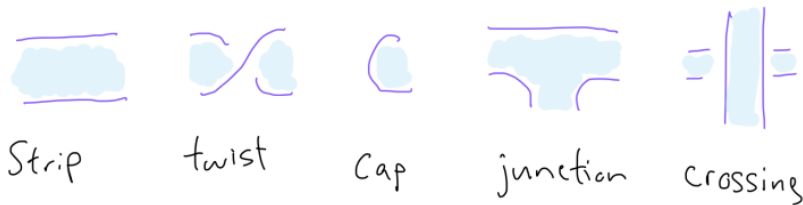


Figure: Building Blocks of Modular Surfaces

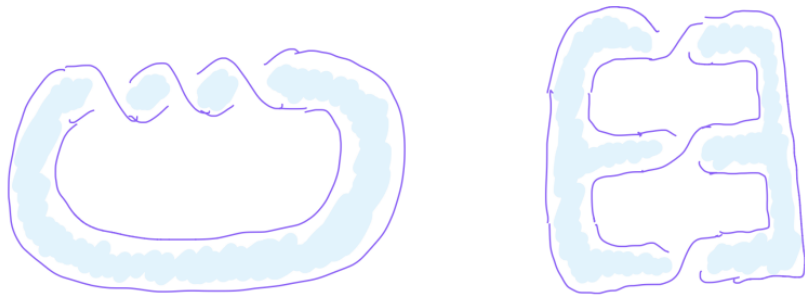


Figure: Two Modular Surfaces Bounded by the Trefoil

A Generalization of Embedded Surfaces

Definition

A surface is immersed instead of embedded if we allow for self intersection. ^a

^aMore rigorously, an immersion is a smooth map such that the derivative is injective at every point

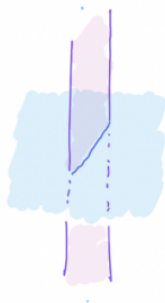


Figure: A ribbon singularity

Immersed Surfaces in Terms of Modular Building Blocks

If we add in one more building block to our previous modular surfaces, we can obtain modular surfaces that are immersed. Such modular surfaces are called **Modular Ribbon Disks**

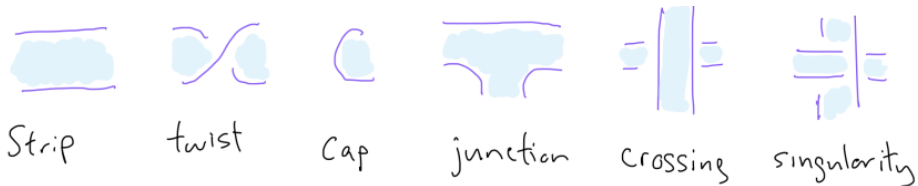


Figure: Ribbon Disk Building Blocks

Ribbon Knots

Definition

A Ribbon Knot is a knot that bounds a ribbon disk. In our framework, we say if a knot K is the boundary of a modular ribbon surface, then K is a ribbon knot.

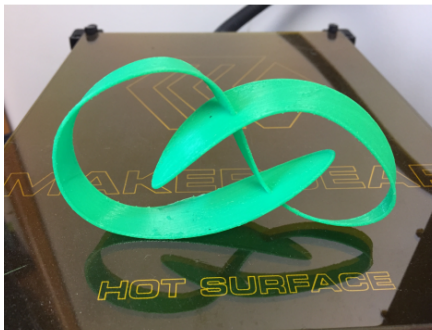


Figure: A 3D Printed Model of The Ribbon Surface Bounded by the Square Knot [Zupan, 2025]

(Briefly) Back to Definitions

Definition

A link is some finite number of circles disjointly embedded in $\mathbb{R}^3$. Think of this as being the union of multiple knots, i.e. $L = K_1 \cup K_2 \cup \dots \cup K_n$

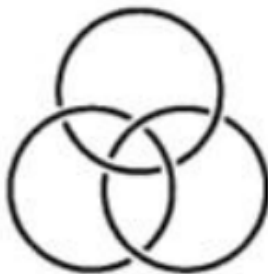


Figure: Borromean Rings

Ribbon Link

Definition

A ribbon link is just a link that bounds a collection of ribbon disks (one for each knot within the link). The disks can have ribbon singularities with themselves or other disks. We call this collection a ribbon surface.



Figure: A ribbon link

The Central Issue

The Problem

There is no general method to determine if a link is ribbon. We can try to find a modular ribbon surface for a link but this is not a straightforward manner, particularly for more complex links. Also as the number of crossings increases so does the number of links.

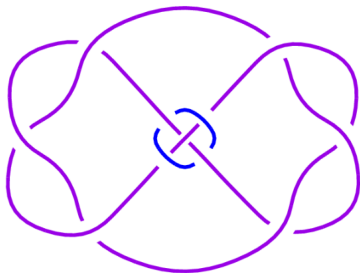


Figure: L11a9

Our Approach

We utilize various **invariants** to reduce the number of possible ribbon links.

Definition

An invariant is some map $\varphi : L \rightarrow \{\text{Polynomials, Elements of Field, Groups, etc}\}$ such that for ambient isotopic links L_1, L_2 , $\varphi(L_1) = \varphi(L_2)$ ^a

^aIn terms of category theory, this can be stated as an invariant is a functor from the category of links to some other category

Examples

The number of crossings of a link, the linking number, the number of components

Reduction of Candidates

There are some known properties about ribbon links in terms of invariants.

Examples (All of these must hold for ribbon links)

Let $L = K_1 \cup K_2 \cup \dots \cup K_n$. If L does not satisfy the following properties, it is not ribbon

- $\det(L)$ is a perfect square (determinant condition)
- Linking matrix consists entirely of zeros
- Jones determinant $\det_V(L)$ is well-defined
- $\det_V(L) \equiv 1 \pmod{8}$
- $\det_V(L) \equiv \det(K_1) \det(K_2) \cdots \det(K_n) \pmod{32}$
- Satisfies the Fox-Milnor condition $\Delta_K^{(n)}(t) = f^{(n)}(t_1, \dots, t_n) f^{(n)}(t_1^{-1}, \dots, t_n^{-1})$

Reduction of Candidates Cont.

We wrote code that filtered out links not satisfying the previous conditions.

NOTE

We can NOT say that links passing these obstructions are ribbon, merely that they might be. Manual checking is required

10 – 11 Crossings

For low crossing links ≤ 11 we used an online database called linkinfo.com.

12 – 14 Crossings

For higher crossing links we generated and extracted data from HTLinkIterator in SnapPy

Results (10 – 11 Crossings)

For links with 10-11 crossings, we were able to find a complete classification.

Theorem

Among all links with 10-11 crossings, the only ribbon links are L_{10n32} and L_{10n36}

We began with approximately 4,000 unique links (up to isotopy), and after application of the invariant obstructions, these were the only two left. Both were verified to be ribbon.

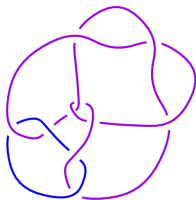


Figure: L_{10n32}

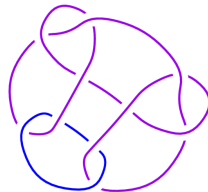


Figure: L_{10n36}

Results (12 – 14 Crossings)

Beginning with 120,000 link diagrams, the filters yielded:

- 92 links passed all obstructions
- 70 of these are previously known ribbon links
- 22 of these are new candidates to check

New Link

We found and verified one new addition to ribbon links, $L13n1091$

Our group was one of three studying ribbon links, and I believe the two other groups were able to verify 3 of the other candidates, in total making 4 new additions to known ribbon links of 12-14 crossings

Summary of Results

Crossings	Candidates	Verified	Notes
≤ 11	2	2	Complete classification
12	18	15	3 newly verified
13	31	28	L13n1091 confirmed
14	43	35	8 under investigation

Table: Ribbon link search results by crossing number

Note on A Constructive Approach to Ribbon Links

Open Problem

Does every ribbon link admit a symmetric union presentation?

So far, every known ribbon link admits a symmetric union presentation, so it seems likely. Using this fact, we took known ribbon links and added symmetric components to make new links and examined if they were ribbon.

Results

All of the links generated in the manner were previously known ribbons, but perhaps this could be a method for higher crossing number links

References



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Vrgras-Lara, Hassan, M. (2017).

Knot energy, complexity, and mobility of knotted polymers.

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Zupan, A. (2025).

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The End

Application of Ribbon Lins

One can construct a 'Ribbon Category' which is a monoidal category equipped with braiding, twist, and duality. It is called ribbon because the morphisms can be drawn as ribbons.

This has applications in braiding of anyons particularly, so the rules for braiding, twisting, etc. of ribbons can be a model for certain phenomena in particle physics.